

Jeffrey Epstein

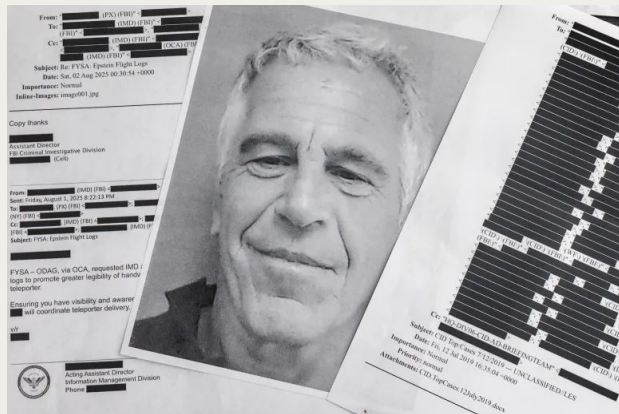
A proposed study by Erica Wang and Sreeyutha Ratala

1

Overview

What are the Epstein Files?

What are the Epstein Files?



- Collection of documents released by the Department of Justice
- Through a series of court investigations about Jeffrey Epstein
- Emails, flight logs, legal records
- Can be considered a dataset of information.

2

Motivation

Why should we care?

Crimes

Search Full Epstein Library

Share



minor

Showing 1 to 10 of 7,726 Results

island

Showing 1 to 10 of 52,828 Results

recruit

Showing 1 to 10 of 1,646 Results.

Political, Economic Influence



From: Jeffrey Epstein <jeevacation@gmail.com>
To: Elon Musk <[REDACTED]>
Subject: Re:
Date: Wed, 11 Sep 2013 20:11:28 +0000

do you think i am retarded, . ? just kidding , there is no one over 25 and all very cute

On Wed, Sep 11, 2013 at 2:04 PM, Elon Musk <[REDACTED]> wrote:

I run and lead product design/engineering for two complicated companies. Moreover, SpaceX is about to launch what is arguably the most advanced rocket in history. Flying to NY to see UN diplomats do nothing would be an unwise use of time.

On Sep 11, 2013, at 5:30 AM, "Jeffrey Epstein" <jeevacation@gmail.com<mailto:jeevacation@gmail.com>>> wrote:

any plans for ny . the opening of the genereal assembly has many interesting people coming to the house

--

Media Literacy



muhalif_silken 1w

They are trying to discredit this important case with videos generated by artificial intelligence. Now it's very clear



294



deni_is_not.here 9w

I mean, it's good that this is getting exposed but Bannon said it himself, give the opposition A LOT of useless information so that they don't know what to protest against and lo and behold: 9500 images and NO LEGAL ACTION, nothing.



149



kamalahahaha.hahahaharris 3w

Where is the evidence of this?

Reply



1



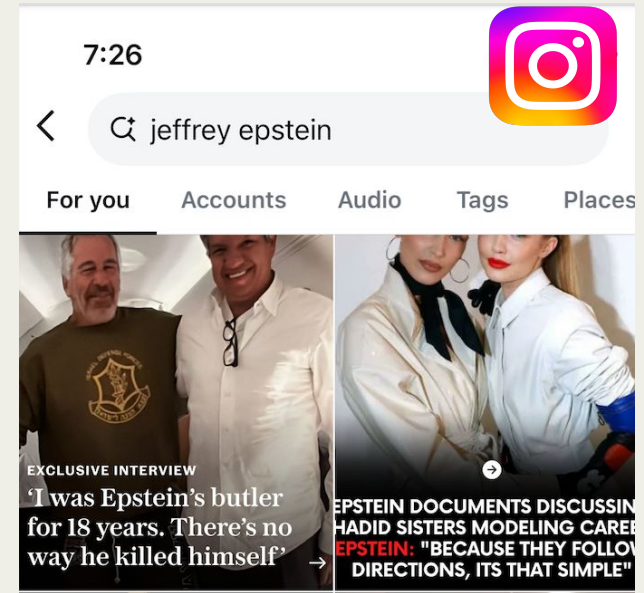
jakeschachte 3w

Which document is this in

Reply



10



- The whole story?
- Or a curated slice

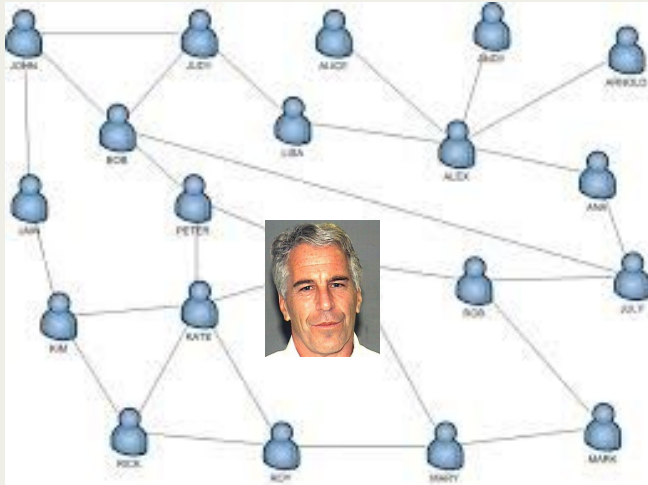
Red herring: a piece of information presented in a way to be distracting

2

Relationship with CS144

How can we connect this to what we have learned?

Email Communication Network



Nodes -> Individuals

Edges -> Mentioned in the same email

We can create an undirected network graph

Using CS144 network concepts below, can we reveal information about the Epstein files?

- Diameter size?
- Centrality?
- Clustering?

3

Our Product

A case study - an investigation of the Epstein File Network through the lens of properties learned in CS144.

Network Centrality : Who Really Mattered?



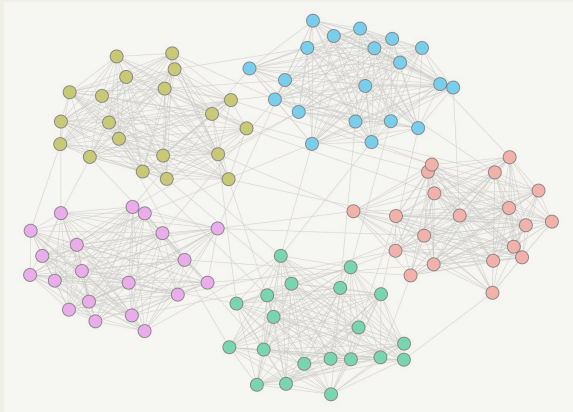
Hypothesis: Epstein served as a communications hub for more powerful individuals.

Tools: Degree, betweenness, and closeness centrality + our own centrality measures.

Output: Nodes of different measures of importance to study.

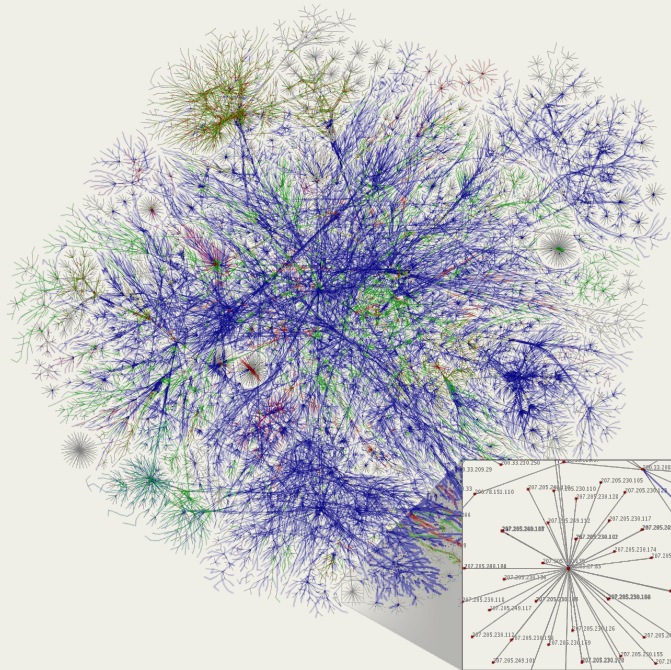
Goal: Who Really Mattered?

Clusters : How do they Relate?



- 1) Identify clusters
- 2) Characterize each cluster
 - a) Research Individuals
 - b) Identify similarities and differences between clusters
- 3) Analyze inter-cluster connectivity
 - a) Bridge nodes
- 4) Summarize insights - what can we learn from the clusters formed around Jeffrey Epstein? Do different clusters serve different purposes?

What was Epstein's Role?



Important: the network graph is built around Epstein.

To investigate his role, we may expand to a larger surrounding network.

- Is Epstein structurally important?
- Unique?
- A heavy-tailed node?

Epstein Files?

Miscellaneous



- The files are highly redacted.
 - Given our network graph, can we make educated guesses on the identity of such individuals?
- Are some nodes more connected than others?
 - Does this graph display a heavy-tailed property?

4

Implementation Details

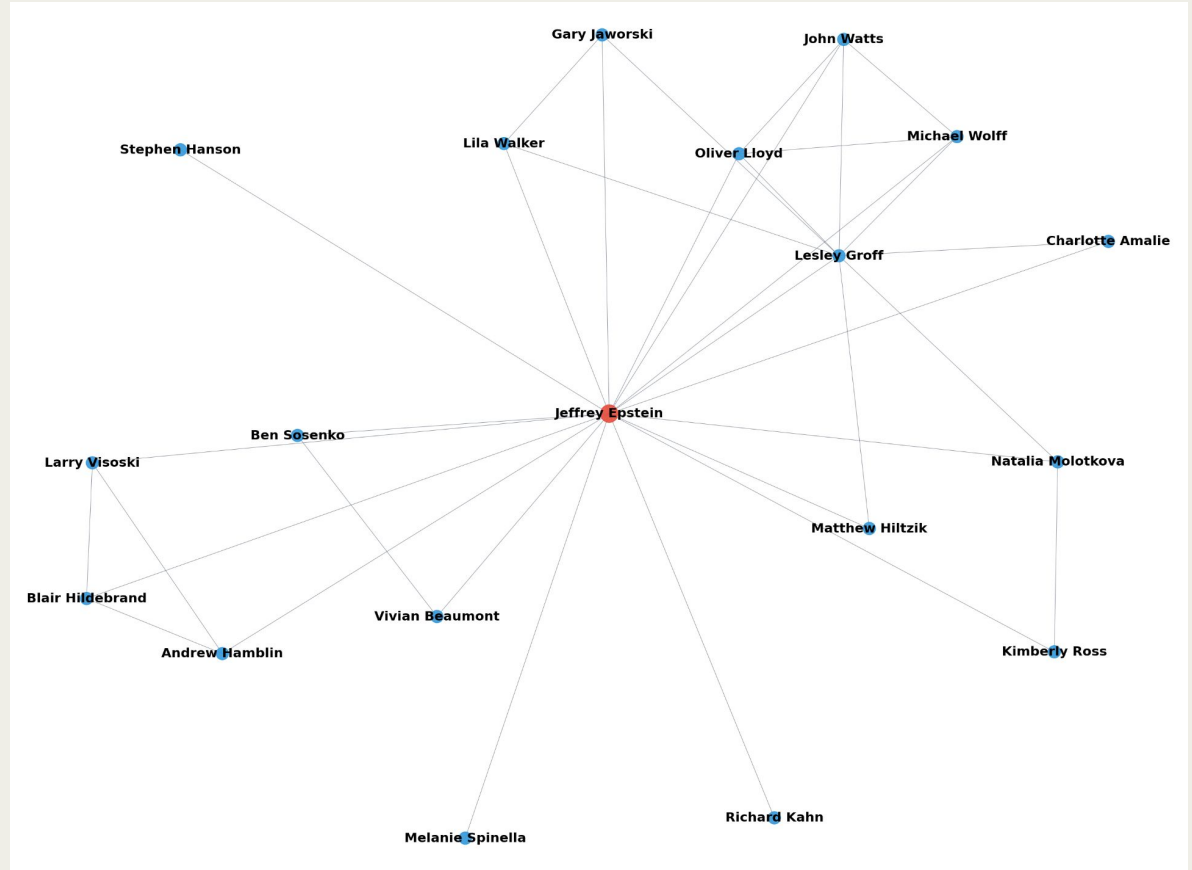
Web crawling

Document Scraping



- Data available via the Department of Justice online
- Network graph obtained via a pdf-scraper
 - Similar to in class
- For potentially a larger scoped graph, we may web-crawl other spaces, such as Wikipedia or the Internet.

Proof of Concept



5

Ending Thoughts

Subjective study, large public impact

Limitations and Concerns

Search Full Epstein Library



mona lisa

Showing 1 to 10 of 17 Results.

- Co-mention != real relationship
 - E.g. Mona Lisa
- Subjective study
 - Network analysis suggests patterns, not definitive conclusions
- Redactions -> incomplete graph

Greater Impact



- Epstein files are a major source of public interest and debate
 - Analysis could impact politics
- Large document collections are difficult to interpret manually
- Our tool provides a computational tool to analyze these relationships
- Network analysis is powerful

Academic Influence

January 10, 2019

Dear Gretchen,

In appreciation of the work being done at the MIT Media Lab, I hereby request that you transfer all eligible funds donated to The Origins Project to the MIT Media Lab whose contact information is as follows:

MIT Media Lab
c/o Marissa Wozniak
77 Massachusetts Avenue, E14-245
Cambridge, MA 02139

With best wishes for the new year.

Sincerely,

Jeffrey Epstein
Enhanced Education



Search Full Epstein Library

harvard

Showing 1 to 10 of 12,170 Results.

Table of contents

1. Overview

2. Motivation

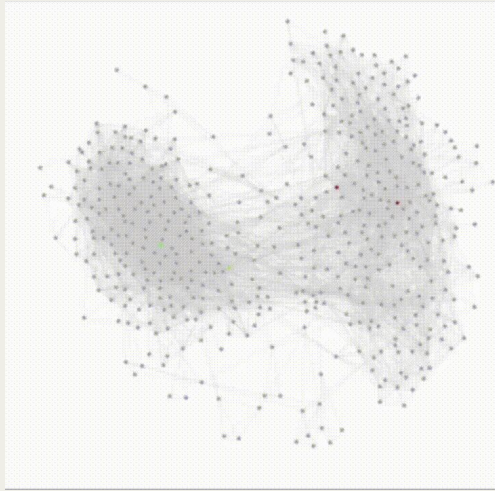
3. Relationship to CS144

4. Our product

5. Implementation details

6. Ending thoughts

Cascades : How Did Epstein's Connections Spread



How did involvement in the files spread over time?

Were some connections never made?

Did some connections cease to exist?

- Cascades
- Update rule?
- Were some individuals more likely to convert to a connection than others?



THRIFTNET

Network-Aware Discovery in
Second-Hand Fashion Ecosystems



OI



MOTIVATION

What's the issue with how second-hand fashion is often discovered now?



WHY IS FINDING THE PERFECT THRIFT ITEM SO HARD?



best thrift stores near pasadena



Full Circle Thrift

4.6 ★★★★★ (106) · Thrift store

Altac

Clos

In-st

Revival Boutique & Thrift

4.4 ★★★★★ (90) · Thrift store

Pasadena Antique Mall

4.4 ★★★★★

309 E Green

Closed · Open

Antiques mall

Rose Bowl Flea Market

4.4 ★★★★★ (1K) · Flea market

1001 Rose Bowl Dr · +1 323-560-7469

Closed · Opens 9 am Sun

Monthly mainstay for new & antique finds

Society of St. Vincent de Paul Los Angel...

4.0 ★★★★★ (933) · Thrift store

Huntington Collection

4.3 ★★★★★ (111) · Second hand store

Hotbox Vintage

4.8 ★★★★★ (30) · The Paseo

Ritz Resale

4.4 ★★★★★

South Pasadena, CA

4.3 ★★★★★ (3.5K) · Shopping mall

1000 E Colorado Blvd · +1 626-795-9100

Remainders Creative Reuse

4.8 ★★★★★ (148) · Thrift store

787 E Washington Blvd suite 101 · +1 626-529-5014

Closed · Opens 10 am

On-site services



food





super unique grey ring

super unique grey ring



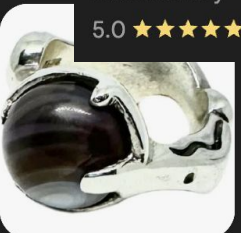
Unique Contemporary
Modern Glass Mar
\$25.00



Ladies Big Silver
Adjustable Glass...



BARIO Neal .98ct Pear
Diamond...



Women's tg-jewellery
by tamara gervasio...
\$390.00



Adjustable Prong Set
with Grey Agate
\$27.18 \$36
Etsy - Seller
Free delivery
5.0 ★★★★★ (1)



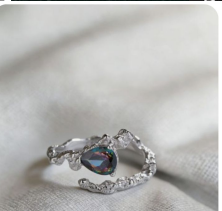
Agate Classic Band
Ring
\$98.00



Moissanite Engage...



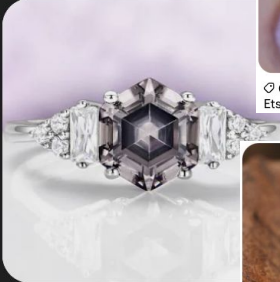
2.10 Carat Salt and Pepper...
Point No Point Studio



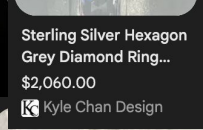
2.10 Carat Salt and Pepper...
Point No Point Studio



Sterling Silver Filigree
Greek Key Women...
\$68.50 \$105
Etsy - Seller
Free delivery
5.0 ★★★★★ (1)



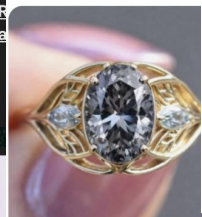
Sterling Silver Hexagon G
Diamond Engagement Rin
\$15.00
Etsy - Aquamarise
Free delivery
9 ★★★★★ (35)



Sterling Silver Hexagon
Grey Diamond Ring...
\$2,060.00
Kyle Chan Design



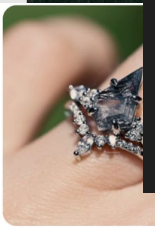
Resin Statement Ring
\$14.00 \$21
Sylca Desig...
Free delivery on \$



Grey Oval Cut Moissanite...
Etsy



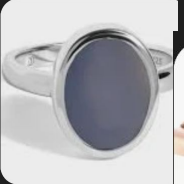
2 Carat Grey Moissanite...
Etsy



Grey Moonstone
Anna Sheffield B
\$2,450.00
Anna Sheffield
Free delivery by T



Oval Gray Diamond Ring in...
Etsy



Crystal Ring - Enfolded Grey
Agate
\$103.00
Moon Magic
Free delivery on \$250+
4.6 ★★★★★ (16)



seree Sesame — Grey Ja
Ring
\$62.00
seree & more
Free delivery on \$75+
3.0 ★★★★★ (2)



Vintage Moissanite
Engagement Ring
\$2,675.00
Rare Earth Jewel
Free delivery



SEVERAL
SONG FILLED
HOURS LATER...



Robynn Good

Yesterday at 7:11PM · 🌐

...

Not weird but definitely wonderful.

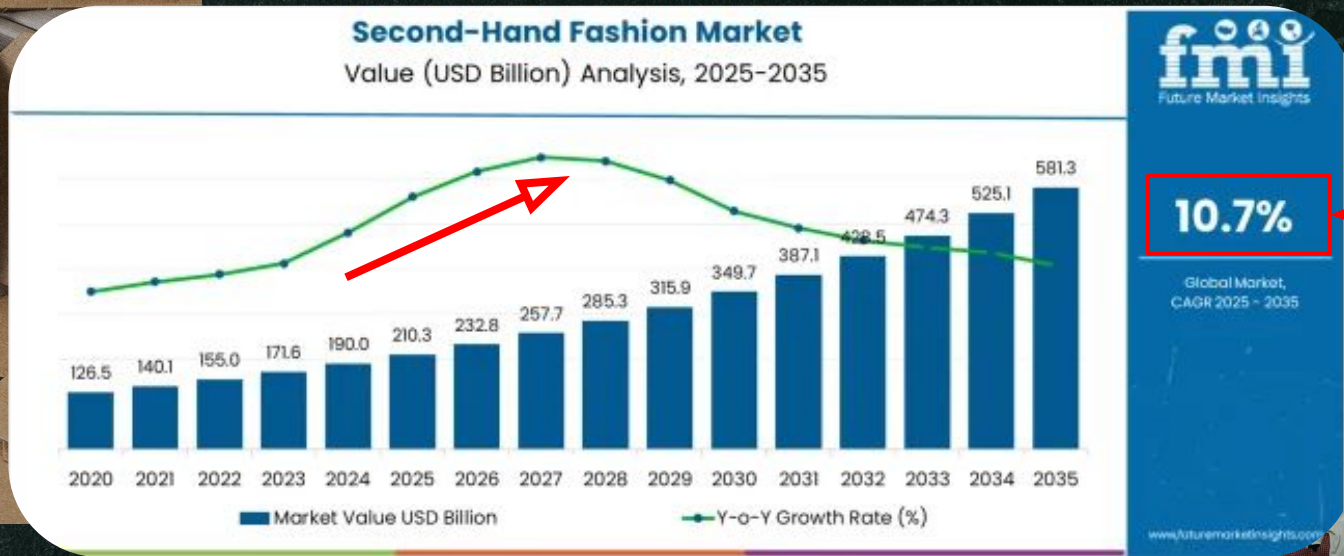
Got this ring from a pawn shop near me called ye old hock shoppe. Real silver and turns out after searching online its "the prince of Lillies" minoan fresco from the greek island of crete. I originally got it cause I loved how unique it is but I love it more now as I follow the greek pantheon in my tradition! Thought I would share!





MARKET GAP

Second Hand Fashion is Growing, Ease of Discovery is not



double-digit
projected
annual
growth

GLOBAL RESALE MARKET GROWING RAPIDLY

\$210B → \$581B





MARKET GAP

Second Hand Fashion is Growing, Ease of Discovery is not

Fragmented; small vendors hidden in otherwise vibrant local ecosystems

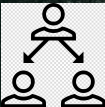
Top Players in Second-Hand Fashion Market*

DIGITAL RESALE PLATFORMS



Digitized individual listings & keyword search

FLEA MARKET VENDORS



LOCAL THRIFT STORES



Decentralized resale ecosystem
Web search, word of mouth, etc.

PAWN SHOPS



VINTAGE BOUTIQUES



INDEPENDENT SELLERS



thriftmeetsboutique ...
Marisa | professional thrifter ★
859 posts 11.1K followers 382 following
Thrift & Consignment Store
hi, I'm Marisa! welcome to my online thrift shop helping the environment 1 thrift at a time 🌱 metro Detroit, MI... more
🔗 linktr.ee/thriftmeetsboutique @thriftmeetsboutique

ETC...



* <https://www.futuremarketinsights.com/reports/second-hand-fashion-market>

02



IDEA & PROOF OF CONCEPT

Thrift Discovery as a Network Problem



Y2K TREND



Influencer (early adopter)

Y2K BUYER CLUSTER



Friend



Friend of friend



Friends adopt y2k



Larger cluster forms

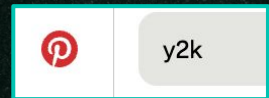


Thrift store stocks y2k IG seller follows trend



Similar thrift/vintage stores follow trend

Y2K SELLER CLUSTER



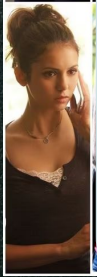
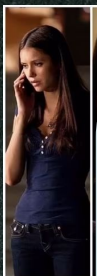
web search only captures trend after network creates it





Y2K TREND

Y2K BUYER CLUSTER

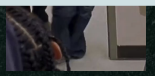


Thrift store stocks y2k IG seller follows trend

Network:

Models how information (i.e. trends) spreads.

Influencer (early adopter)



Friends adopt y2k



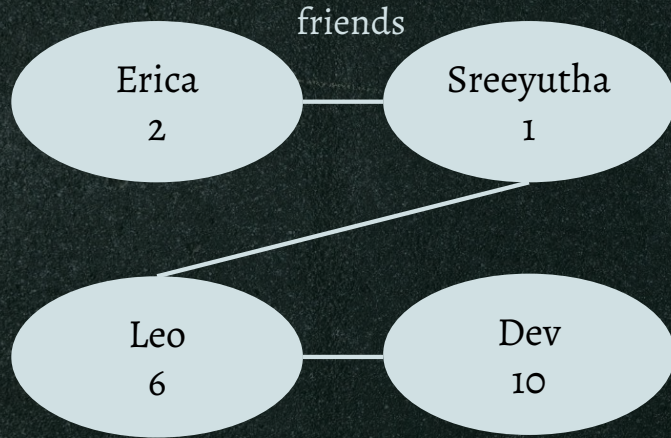
y2k

web search only captures trend after network creates it





THRIFTNET: BUYER NETWORK



user_id	name	age	home_area	user_type	style_tags	notes
u01	Sreeyutha	21	Pasadena	Explorer	Y2K / Vintage / Streetwear	mix of trendy + unique finds
u02	Erica	21	Pasadena	Trend Focused	Y2K / Coquette / Streetwear	knows what's popular, follows fashion content
u06	Leo	22	Echo Park	Explorer	Streetwear / Sneakers	streetwear oriented, resale-aware
u10	Dev	22	DTLA	Mixer	Vintage / Fusion	strong cross-style exploration

When an individual joins the ThriftNet app, they indicate their friends.

1 undirected edge connects 2 buyers that are friends.

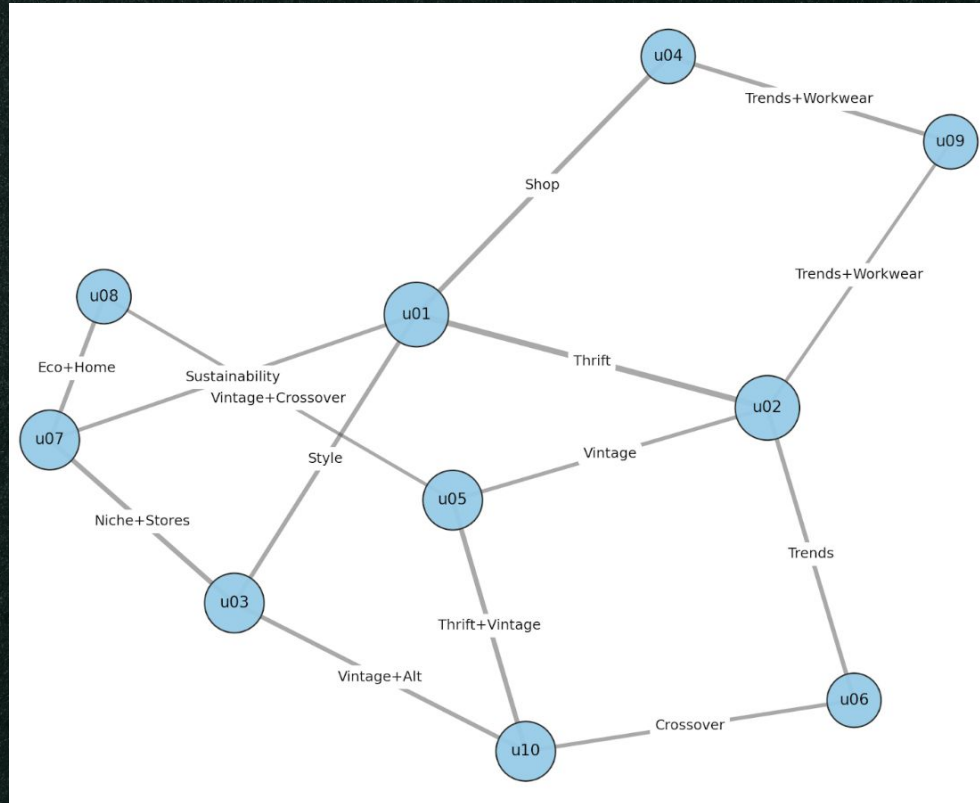
Buyer Profiles
(metadata)





THRIFTNET: BUYER NETWORK

Proof of concept



Extend to more buyers for project





THRIFTNET: SELLER NETWORK

Sellers added using:



THRIFT MAPS



local stores in Pasadena, google maps, yelp reviews, instagram pages, buyer on ThriftNet can add sellers



seller_id	name	seller_type	area	has_physical_location	notes
s01	Revival Boutique	Thrift Store	Pasadena	Yes	Clean/organized; wide variety; community-focused nonprofit
s02	2nd STREET	Resale Store	Old Town Pasadena	Yes	Designer/vintage/trendy; younger clientele; strong brand presence
s06	MeowMeowz	Vintage Thrift	East Pasadena	Yes	Punk/goth niche; live music weekends; quirky inventory

Edge if: same area, same style type, OR $\geq T=1$ shared tags*

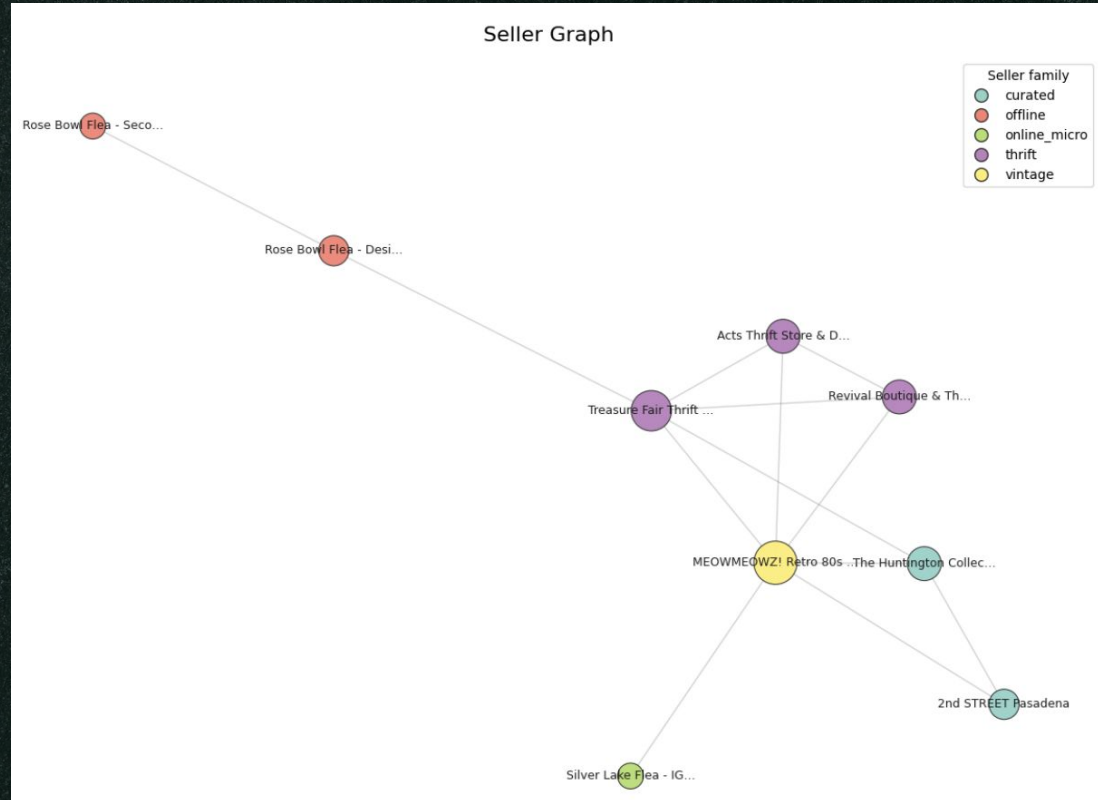
Seller Profiles
(metadata)





THRIFTNET: SELLER NETWORK

Proof of concept



Extend to more
sellers for project
using:



THRIFT MAPS

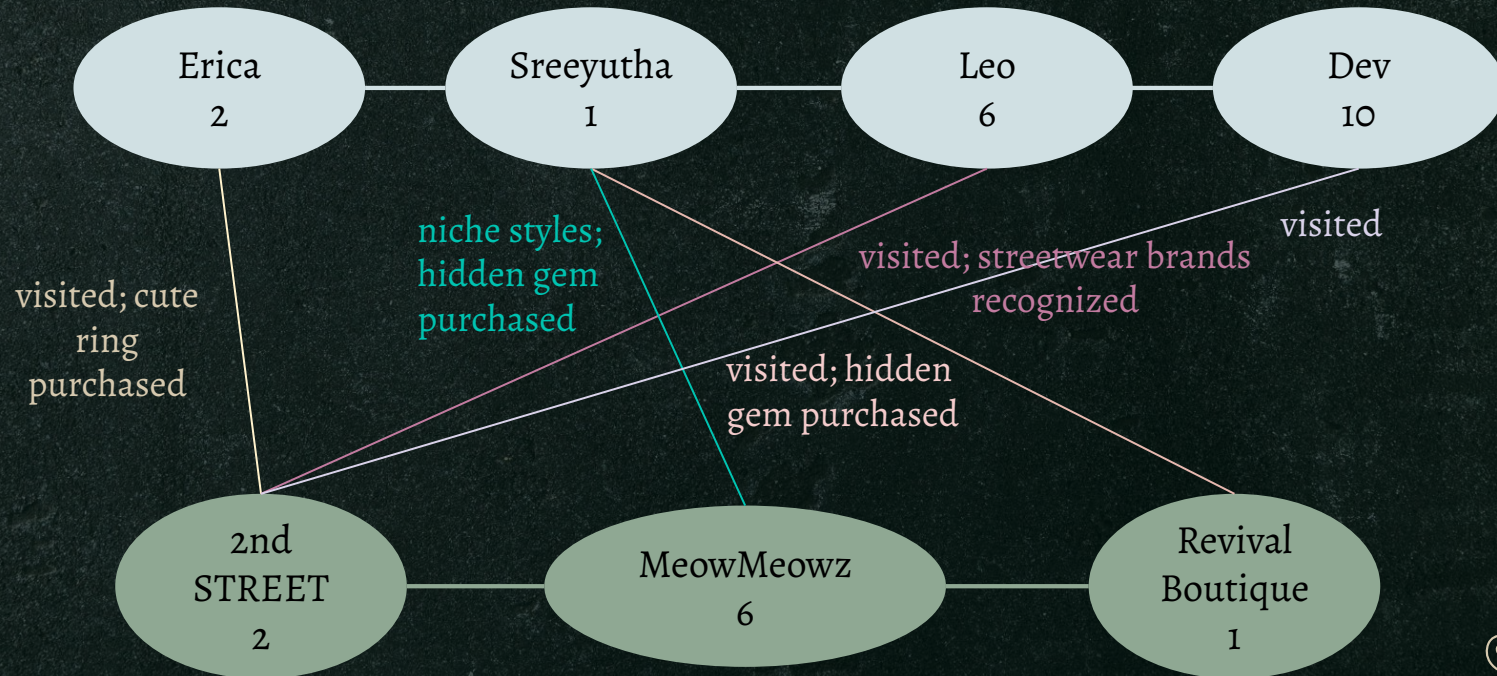




THRIFTNET: BUYER-SELLER NETWORK

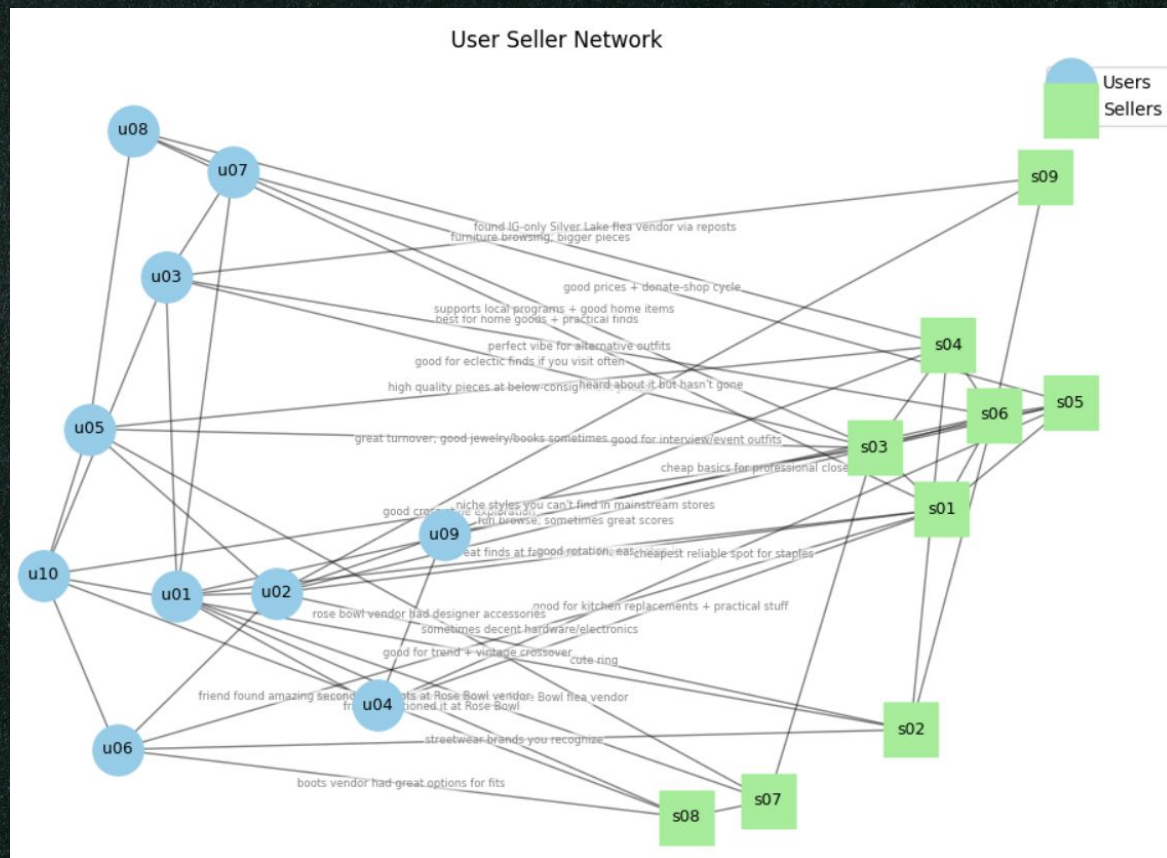
buyer-seller edge = store visited or purchase made

edge metadata: indicates quality of connection





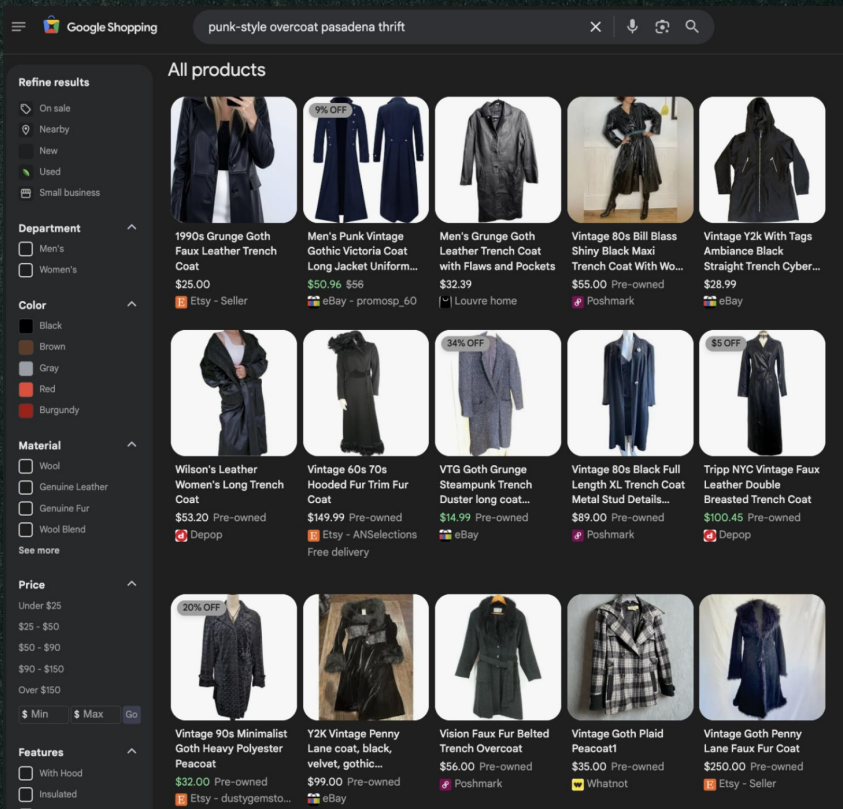
Proof of concept



Extended for
project



EX: ALI WANTS A PUNK-STYLE OVERCOAT FOR GREAT PRICE



TRY I: WEB SEARCH USING KEYWORDS

- Results from online stores: Depop, Etsy, Poshmark, Ebay
- Misses local stores like Meowmeows, which is a great find in this case
 - no online website (hidden digitally)
- Ali isn't satisfied as these don't match her style
 - She saw her friend wear something similar, and was **influenced**



TRY 2: GRAPH TRAVERSAL FOR THRIFT DISCOVERY

Ali is searching for a punk-style overcoat in pasadena to thrift

Ali
8

Erica
2

Sreeyutha
1

Leo
6

Dev
10

visited; cute
ring
purchased

niche styles;
hidden gem
purchased

visited; streetwear brands
recognized

visited

visited

2nd
STREET
2

MeowMeowz
6

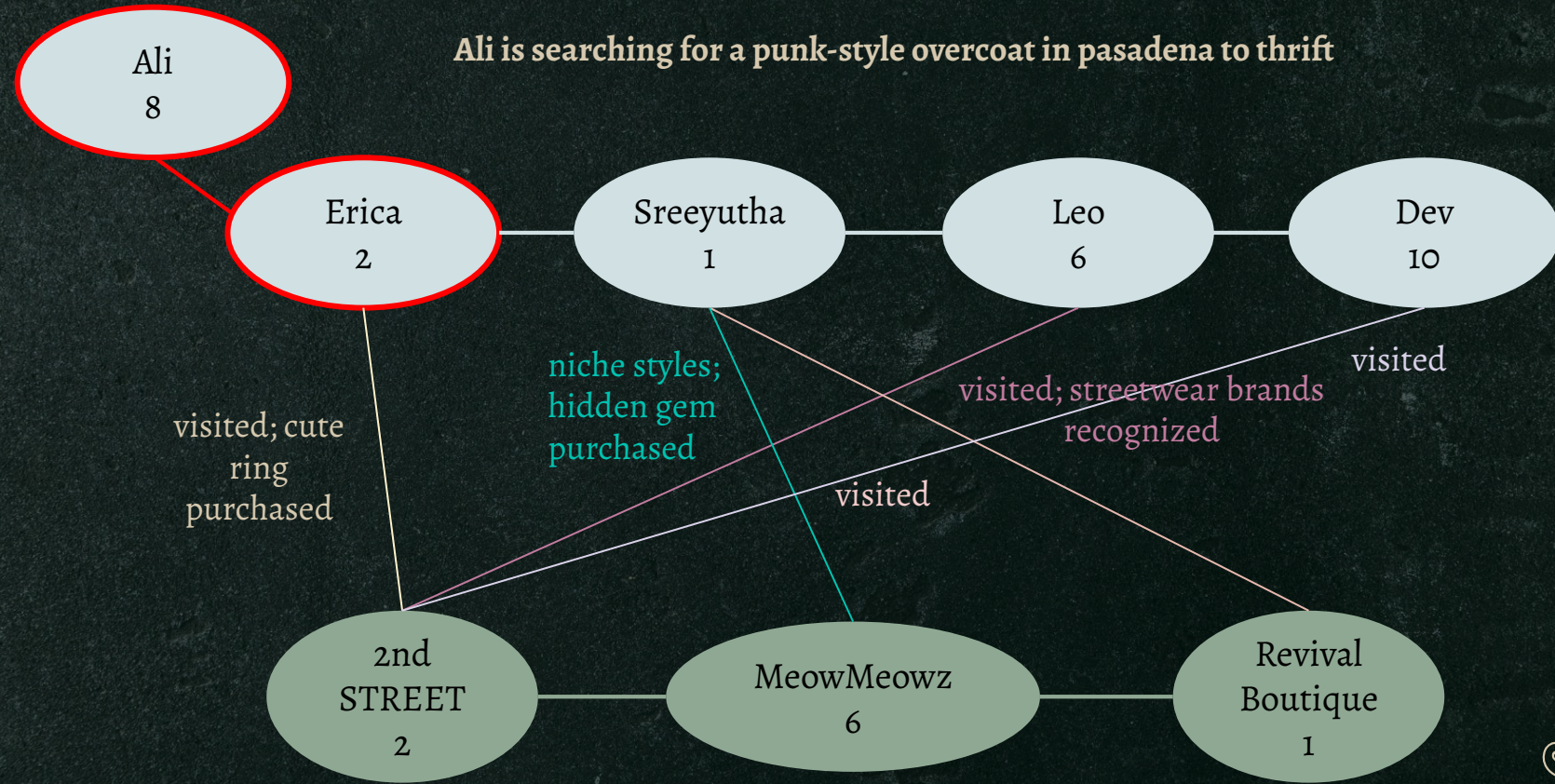
Revival
Boutique
1





TRY 2: GRAPH TRAVERSAL FOR THRIFT DISCOVERY

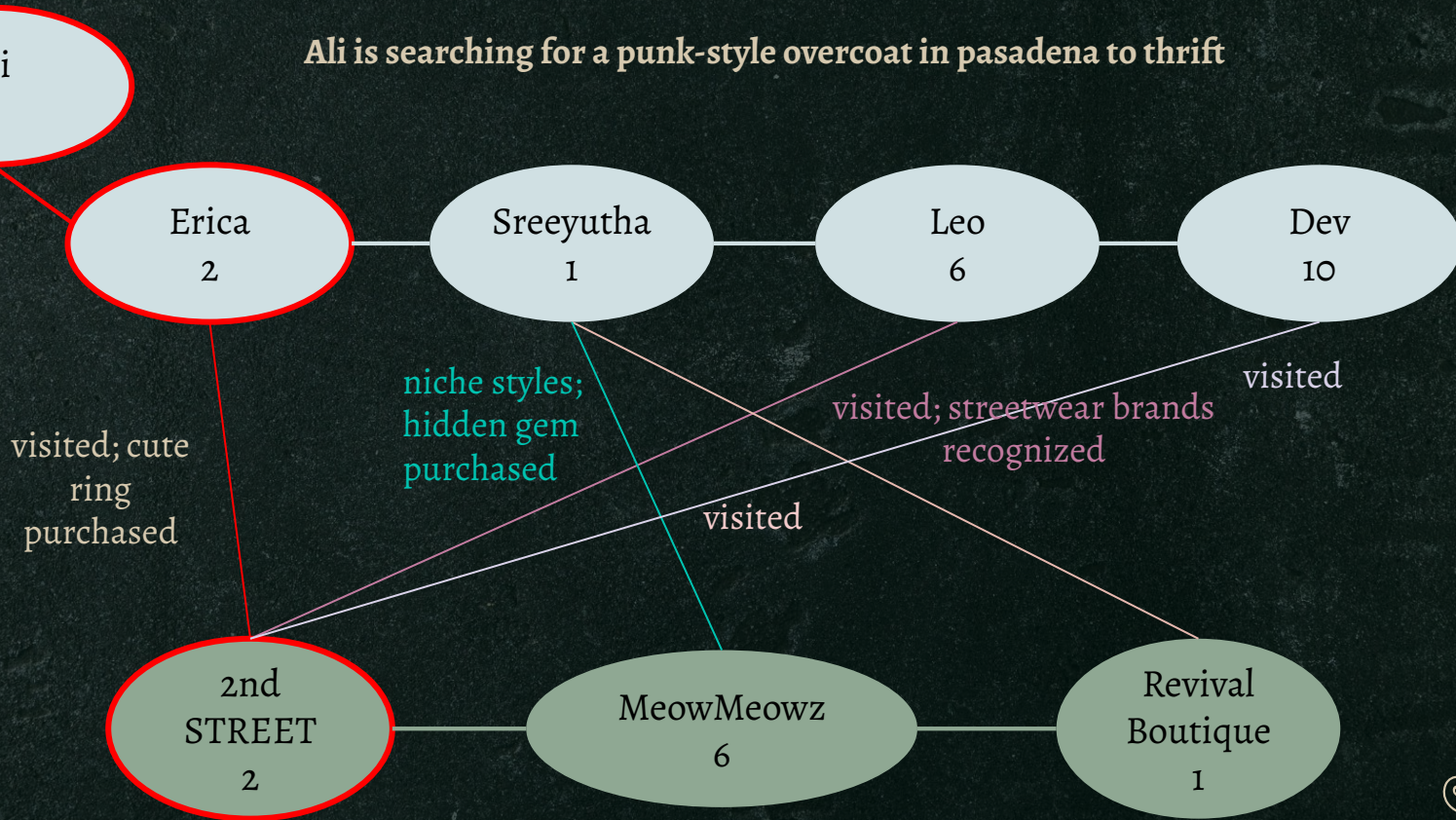
Ali is searching for a punk-style overcoat in pasadena to thrift





TRY 2: GRAPH TRAVERSAL FOR THRIFT DISCOVERY

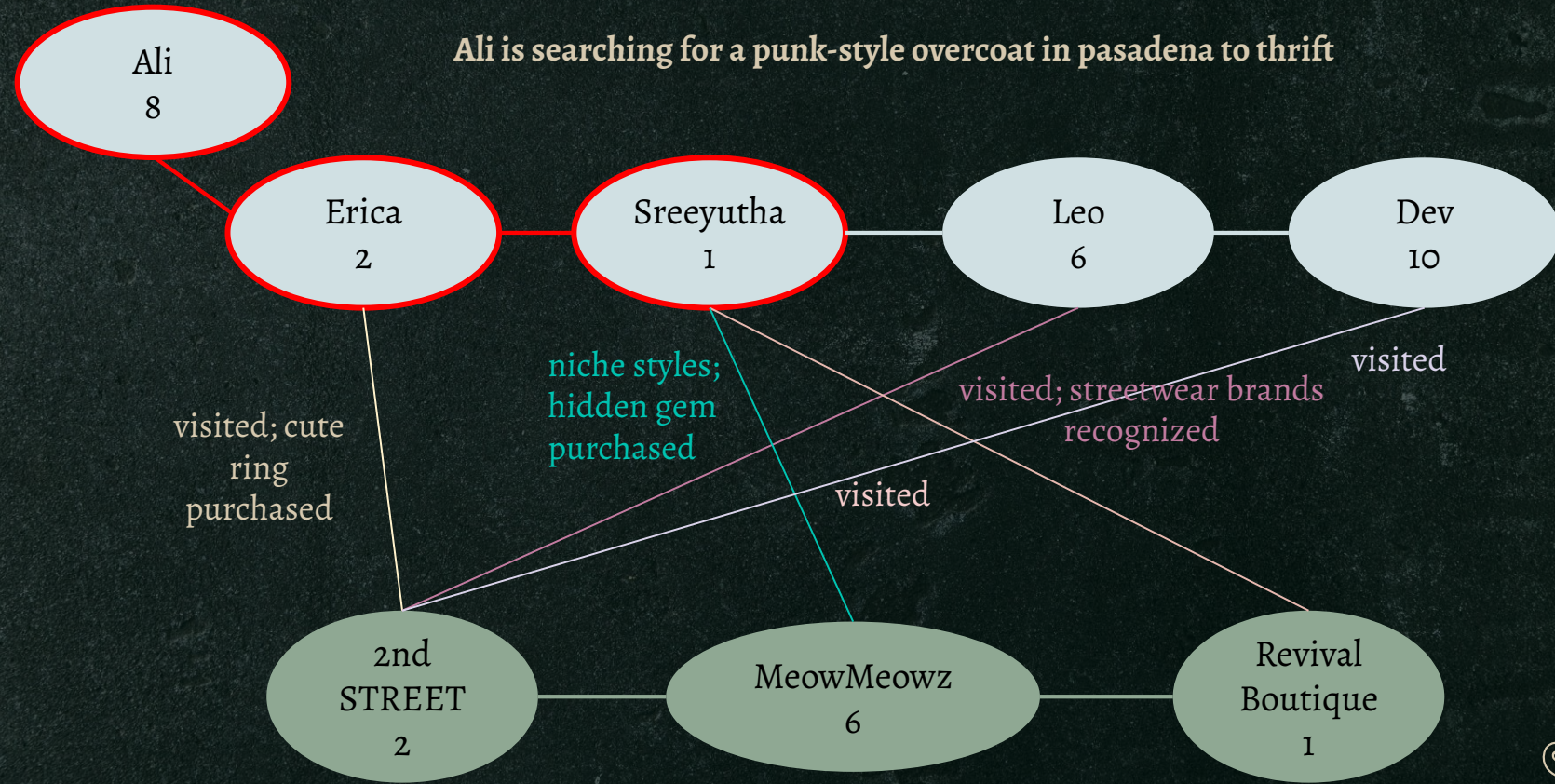
Ali is searching for a punk-style overcoat in pasadena to thrift





TRY 2: GRAPH TRAVERSAL FOR THRIFT DISCOVERY

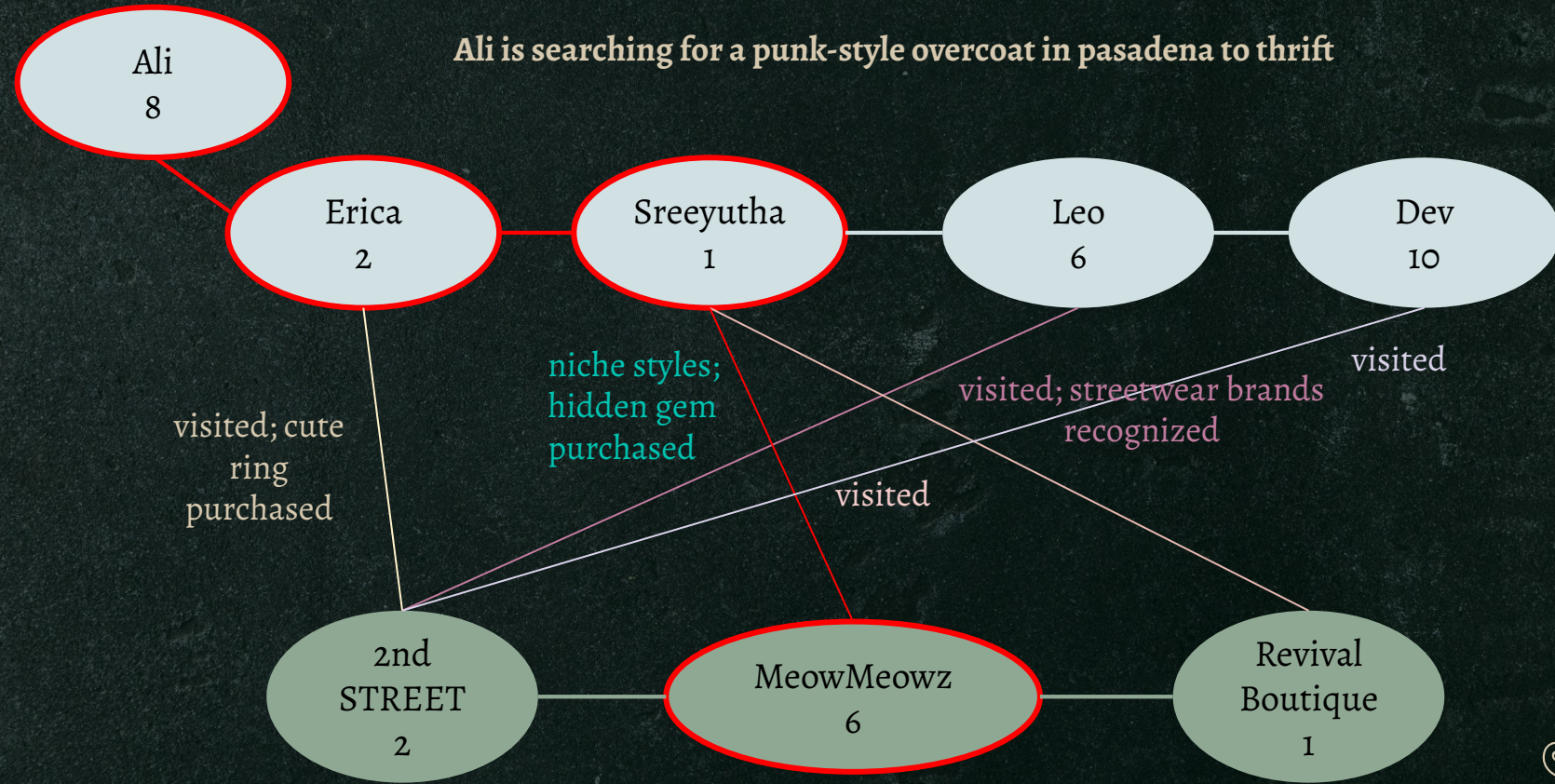
Ali is searching for a punk-style overcoat in pasadena to thrift





TRY 2: GRAPH TRAVERSAL FOR THRIFT DISCOVERY

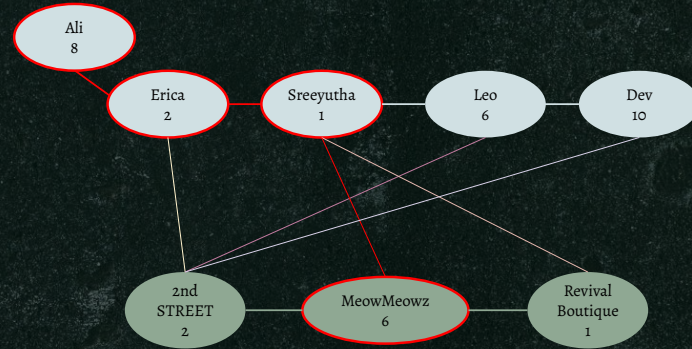
Ali is searching for a punk-style overcoat in pasadena to thrift





ALI VISITS MEOWMEOWZ

She finds...



This store is close to her in the network, and has the exact vibe she's going for! (suboptimal case)

Found through a short path in network

A nearby niche seller surfaced through social connections



03



END PRODUCT

A New Application & Second Hand Fashion Insights

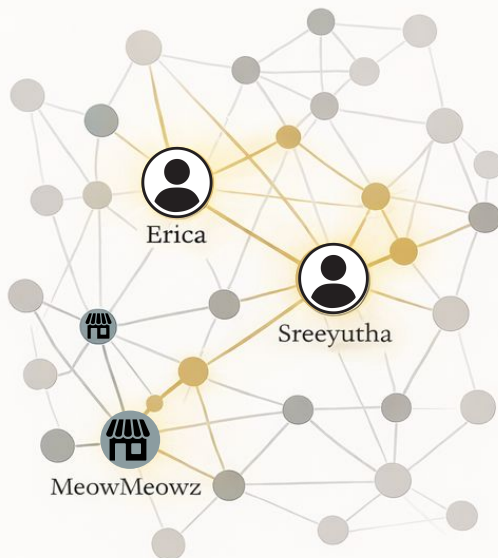


Erica
Pasadena

- Trend Focused
- Y2K / Coquette / Streetwear

FILTERS

traverse graph



About this map >

← BACK

🔍 ZOOM

🌟 EXPLORE

Recommended Thrift Stores



MeowMeowz

Punk / vintage hidden gem



Revival Boutique

Curated resale / nonprofit



rec2



rec3



[VIEW ALL](#)

Explore whether ThriftNet (incorporating social structure into second-hand fashion) can improve thrift discovery experiences for buyers and sellers (often decentralized) in LA



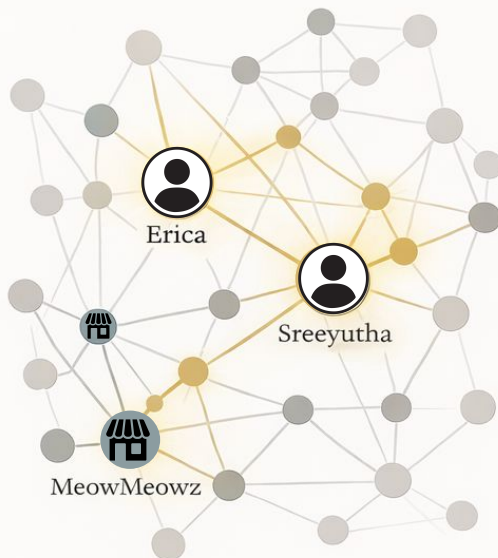
Erica
Pasadena

- Trend Focused
- Y2K / Coquette / Streetwear

FILTERS 



traverse graph



About this map >

← BACK

🔍 ZOOM

🌟 EXPLORE

Recommended Thrift Stores



MeowMeowz

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Revival Boutique

Curated resale / nonprofit



rec2



rec3



VIEW ALL



Erica
Pasadena

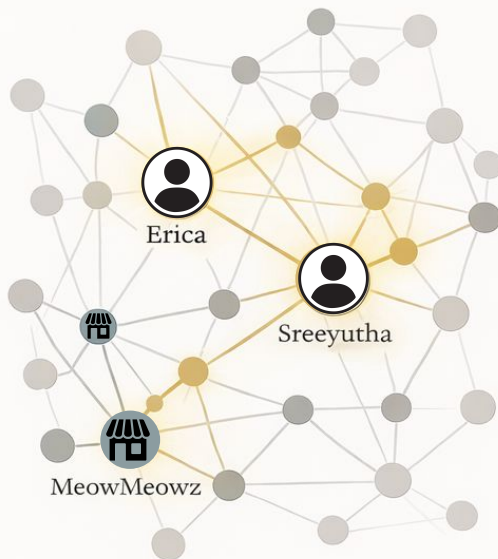
- Trend Focused
- Y2K / Coquette / Streetwear

FILTERS



traverse graph

buyers have option to
search through graph
themselves



About this map >

← BACK

🔍 ZOOM

🌟 EXPLORE

Recommended Thrift Stores



MeowMeowz

Punk / vintage hidden gem



Revival Boutique

Curated resale / nonprofit



rec2



rec3



VIEW ALL



Erica
Pasadena

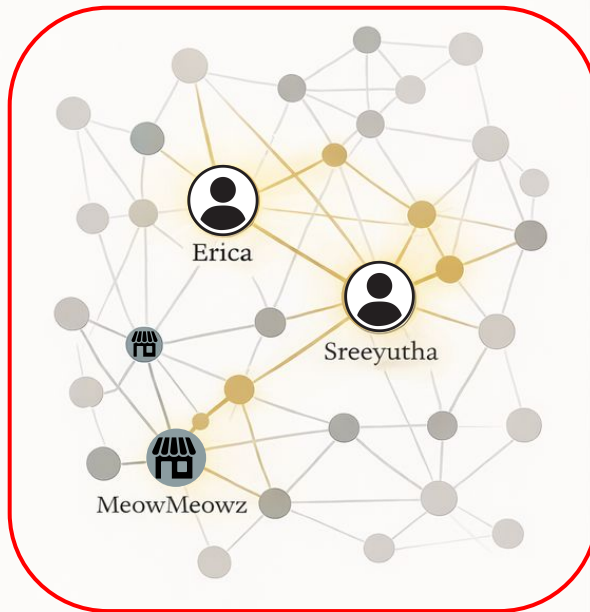
- Trend Focused
- Y2K / Coquette / Streetwear

FILTERS 



 traverse graph

thrift network (dynamic - changes as
buyers & sellers are added to network)



About this map >

← BACK

🔍 ZOOM

🌟 EXPLORE

Recommended Thrift Stores



MeowMeowz

Punk / vintage hidden gem



Revival Boutique

Curated resale / nonprofit



rec2



rec3



VIEW ALL



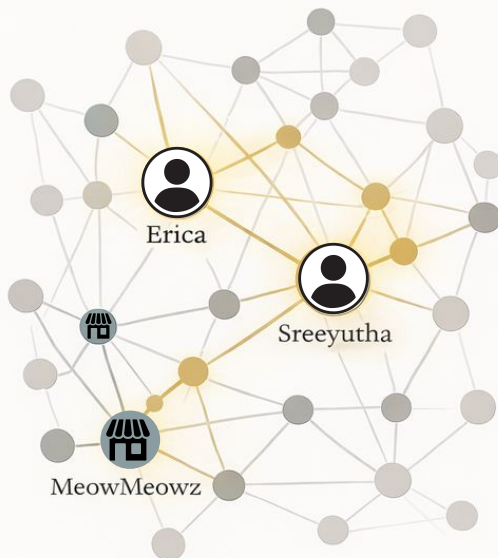
Erica
Pasadena

- Trend Focused
- Y2K / Coquette / Streetwear

FILTERS 



 traverse graph



About this map >

← BACK

🔍 ZOOM

🌟 EXPLORE

Recommended Thrift Stores



MeowMeowz

Punk / vintage hidden gem



Revival Boutique

Curated resale / nonprofit



rec2



rec3



VIEW ALL



INSIGHTS FROM TRIFT NETWORK

Can network structure (social) improve thrift seller discovery using centrality, clustering, and graph traversal methods?



CENTRALITY

**Which sellers become hubs?
(Ranking sellers)**

Pagerank, degree centrality



CLUSTERING

**Which sellers/buyers form
style communities?**

Spectral clustering



TRAVERSAL

**How do buyers reach sellers?
How do “hidden gem” sellers
spread socially?**

Shortest paths BFS/DFS,
neighborhood expansion





THANKS!

Questions?

